

WEEK 7

Name - Ch vinay

Roll No - A23126510015

Branch - CSE A

For your first Jenkins setup, choose “Install suggested plugins.” It is the easiest option.

What to do now

1. Click Install suggested plugins.
2. Jenkins will automatically install the commonly required plugins.
3. Wait for the installation to complete. It may take a few minutes.
4. After that, Jenkins will show the Create First Admin User screen.
5. Enter:
 - o **Username**: admin (or your preferred username)
 - o **Password**: create a password you can remember
 - o **Full name**: your name
 - o **Email**: your email address
6. Click **Save and Continue**.
7. On the next screen, **click Save and Finish**.
8. Click **Start using Jenkins**.

For your Java + Maven project

Suggested plugins are enough for now.

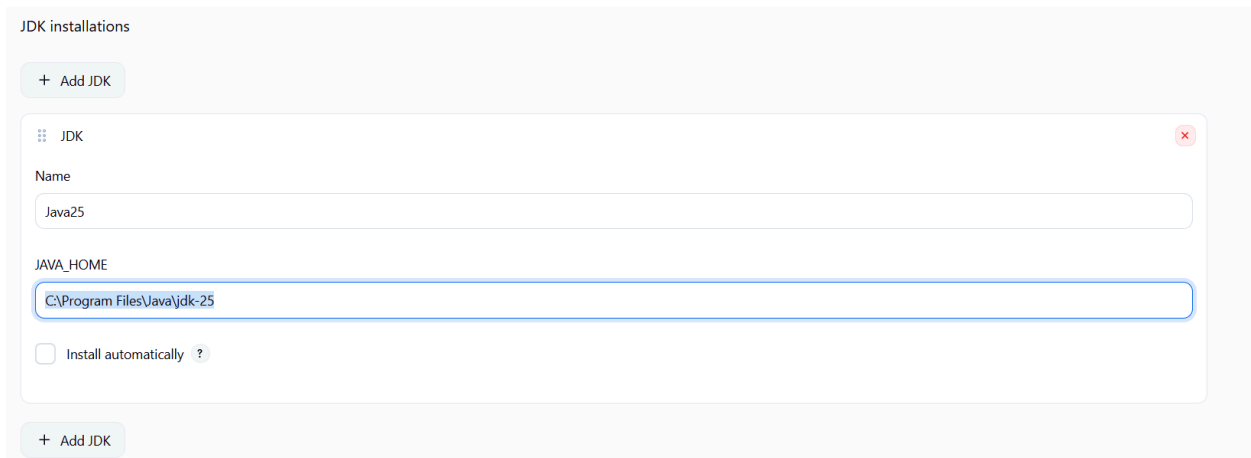
Perfect. You are on the correct **Tools** page.

We only need to configure **JDK, Git, and Maven** for your Java + Maven Jenkins project.

1. Configure JDK

Name: Java25

If there is an **Installation path** field, enter the location of your **JDK 25**, for example:
C:\Program Files\Java\jdk-25



JDK installations

+ Add JDK

JDK

Name

Java25

JAVA_HOME

C:\Program Files\Java\jdk-25

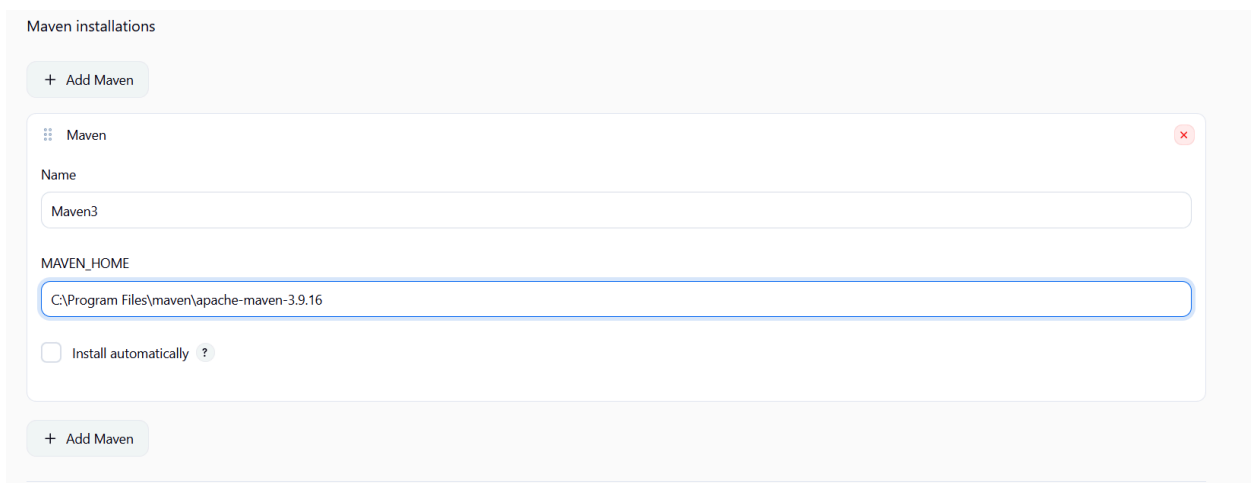
☐ Install automatically ?

+ Add JDK

2. Configure Maven

Name: Maven3

C:\Program Files\maven\apache-maven-3.9.16



Maven installations

+ Add Maven

Maven

Name

Maven3

MAVEN_HOME

C:\Program Files\maven\apache-maven-3.9.16

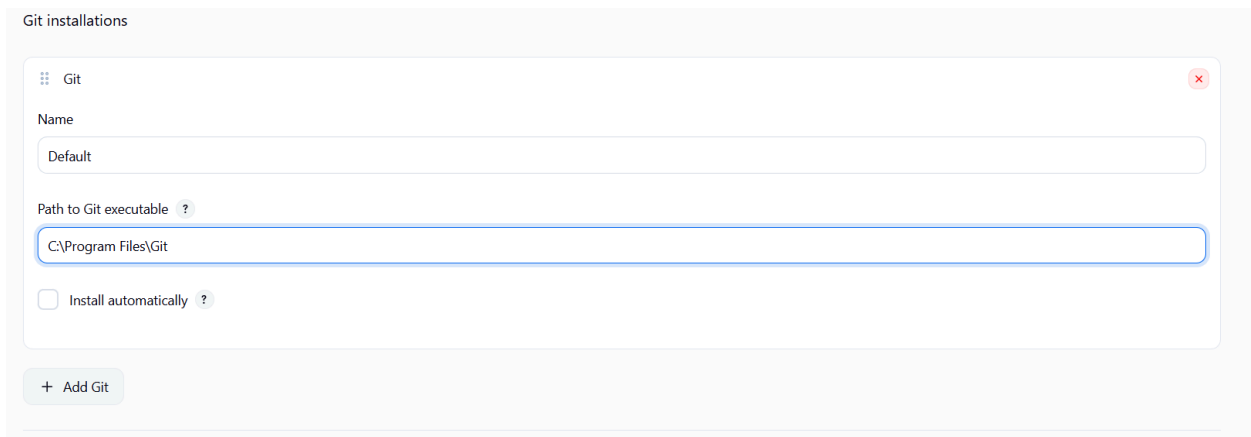
☐ Install automatically ?

+ Add Maven

3. Git

Name: default

C:\Program Files\Git



Git installations

Name	Path to Git executable
Default	C:\Program Files\Git

☐ Install automatically

+ Add Git

Step 1: Create the Java Maven project

Create the following folder structure on your computer:

```
mkdir C:\JenkinsDemo\src\main\java\com\example
```

```
tree C:\JenkinsDemo
```

```
PS C:\Users\vinay> mkdir C:\JenkinsDemo\src\main\java\com\example

Directory: C:\JenkinsDemo\src\main\java\com

Mode                LastWriteTime         Length Name
----                -
d-----          26-08-2026   19:03         example

PS C:\Users\vinay> tree C:\JenkinsDemo
Folder PATH listing for volume OS
Volume serial number is 000002C8 3A87:10A5
C:\JENKINSDEMO
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   └── example
```

App.java

Put this inside App.java:

```
src > main > java > com > example > App.java > App
1 package com.example;
2
3 public class App {
4     public static void main(String[] args) {
5         System.out.println(x: "Hello world from Jenkins!");
6     }
7 }
```

Pom.xml

Put this inside pom.xml:

```
JenkinsDemo Welcome X App.java X pom.xml X
pom.xml
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4     https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>com.example</groupId>
9     <artifactId>jenkins-demo</artifactId>
10    <version>1.0</version>
11
12    <properties>
13        <maven.compiler.source>21</maven.compiler.source>
14        <maven.compiler.target>21</maven.compiler.target>
15    </properties>
16
17    <build>
18        <plugins>
19            <plugin>
20                <groupId>org.codehaus.mojo</groupId>
21                <artifactId>exec-maven-plugin</artifactId>
22                <version>3.5.0</version>
23                <configuration>
24                    <mainClass>com.example.App</mainClass>
25                </configuration>
26            </plugin>
27        </plugins>
28    </build>
29
30 </project>
```

Step 2: Test the project outside Jenkins

Open Command Prompt:

```
cd C:\JenkinsDemo
```

Then run:

```
$env:Path += ";C:\Program Files\maven\apache-maven-3.9.16\bin"
```

```
mvn clean compile
```

```
PS C:\JenkinsDemo> mvn clean compile
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compilers/2.16.2/plexus-compilers-2.16.2.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compilers/2.16.2/plexus-compilers-2.16.2.pom (1.6 kB at 44
/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.9.1/asm-9.9.1.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.9.1/asm-9.9.1.jar (126 kB at 2.1 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.5.2/plexus-java-1.5.2.jar
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.2.0/qdox-2.2.0.jar
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.16.2/plexus-compiler-api-2.16.2.jar
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.16.2/plexus-compiler-manager-2.16.2.jar
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/2.16.2/plexus-compiler-javac-2.16.2.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/2.16.2/plexus-compiler-javac-2.16.2.jar (30
at 854 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.16.2/plexus-compiler-manager-2.16.2.jar
.2 kB at 150 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.16.2/plexus-compiler-api-2.16.2.jar (29 kB a
765 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.5.2/plexus-java-1.5.2.jar (57 kB at 1.2 MB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.2.0/qdox-2.2.0.jar (353 kB at 3.3 MB/s)
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 21
not setting the location of system modules may lead to class files that cannot run on JDK 21
--release 21 is recommended instead of -source 21 -target 21 because it sets the location of system modules automatically
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.044 s
[INFO] Finished at: 2026-08-26T19:25:18+05:30
[INFO] -----
```

After successful compilation, run:

```
mvn exec:java
```

You should get:

```

PS C:\JenkinsDemo> mvn exec:java
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.4.0/commons-exec-1.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.4.0/commons-exec-1.4.0.pom (9.5 kB at 232 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/65/commons-parent-65.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/65/commons-parent-65.pom (78 kB at 1.7 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.7.1/asm-9.7.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.7.1/asm-9.7.1.pom (2.4 kB at 62 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.7.1/asm-commons-9.7.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.7.1/asm-commons-9.7.1.pom (2.8 kB at 74 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.7.1/asm-tree-9.7.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.7.1/asm-tree-9.7.1.pom (2.6 kB at 68 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.4.1/maven-resolver-util-1.4.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.4.1/maven-resolver-util-1.4.1.jar (2.9 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.4.1/maven-resolver-api-1.4.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.4.0/commons-exec-1.4.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.7.1/asm-commons-9.7.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.7.1/asm-9.7.1.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.7.1/asm-tree-9.7.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.7.1/asm-9.7.1.jar (126 kB at 2.7 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.7.1/asm-tree-9.7.1.jar (52 kB at 416 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.4.0/commons-exec-1.4.0.jar (66 kB at 519 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-commons/9.7.1/asm-commons-9.7.1.jar (73 kB at 532 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.4.1/maven-resolver-api-1.4.1.jar (1.0 MB/s)

Hello World from Jenkins!

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.676 s
[INFO] Finished at: 2026-08-26T19:29:07+05:30
[INFO] -----

```

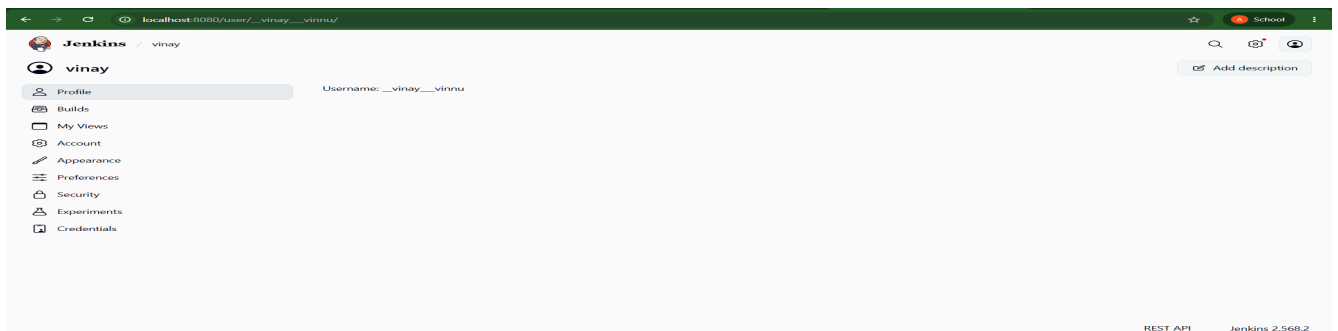
This confirms that your Java + Maven project works before involving Jenkins.

Step 3: Create a Jenkins Job

Open:

<http://localhost:8080>

On the Jenkins Dashboard:



Click → New Item

Enter:

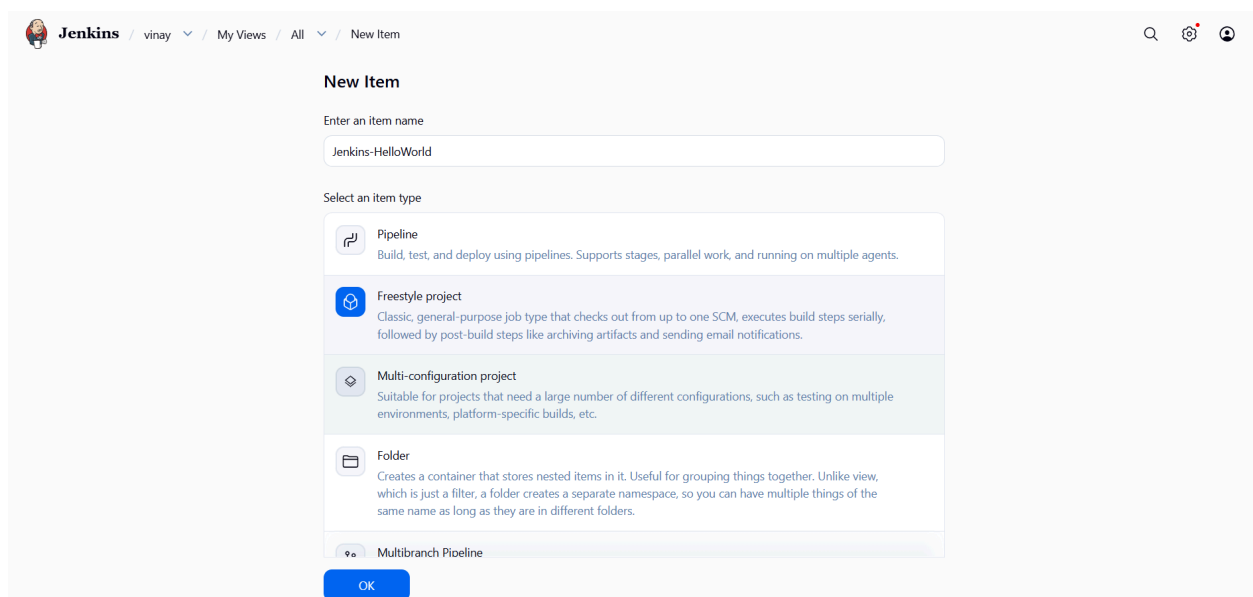
Jenkins-HelloWorld

Select:

Freestyle project

Then click:

OK



Step 4: Configure the job

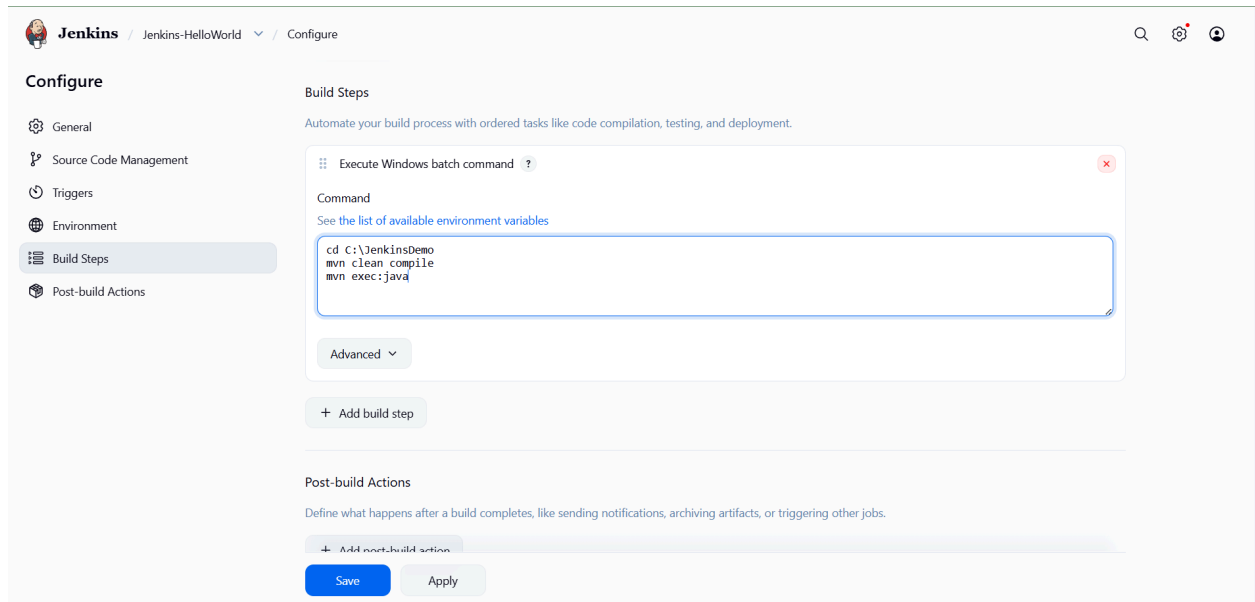
Scroll down to Build Steps.

Click:

Add build step → Execute Windows batch command

Enter:

```
cd C:\JenkinsDemo
mvn clean compile
mvn exec:java
```



So Jenkins will execute:

C:\JenkinsDemo



mvn clean compile



mvn exec:java



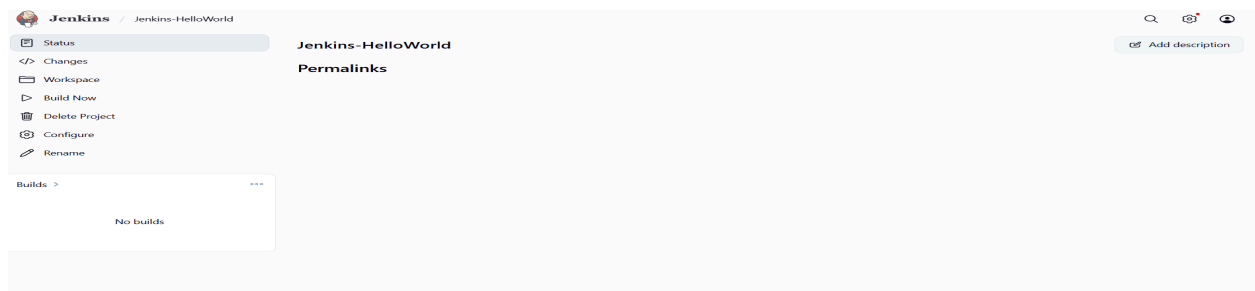
Hello World from Jenkins!

Step 5: Save

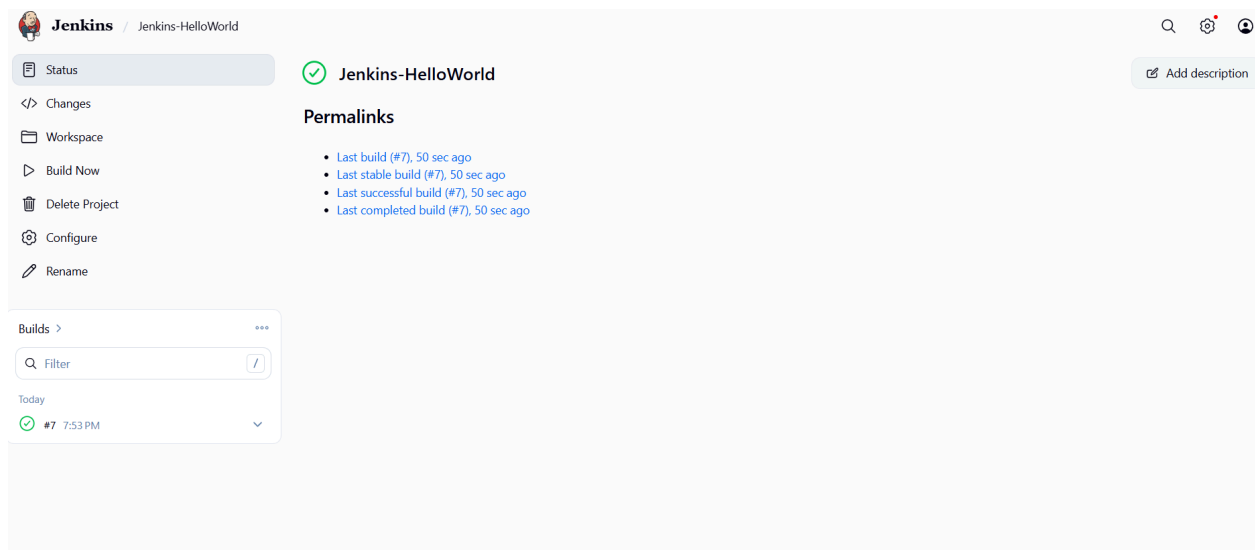
Click:

Save

You will return to the Jenkins project page.

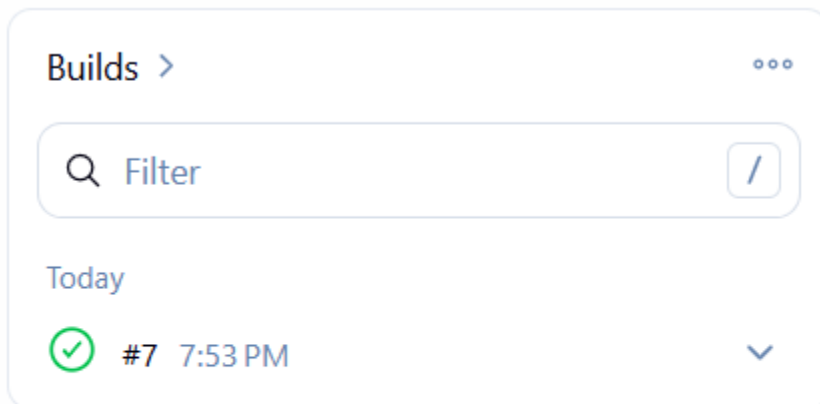


Click:
Build Now

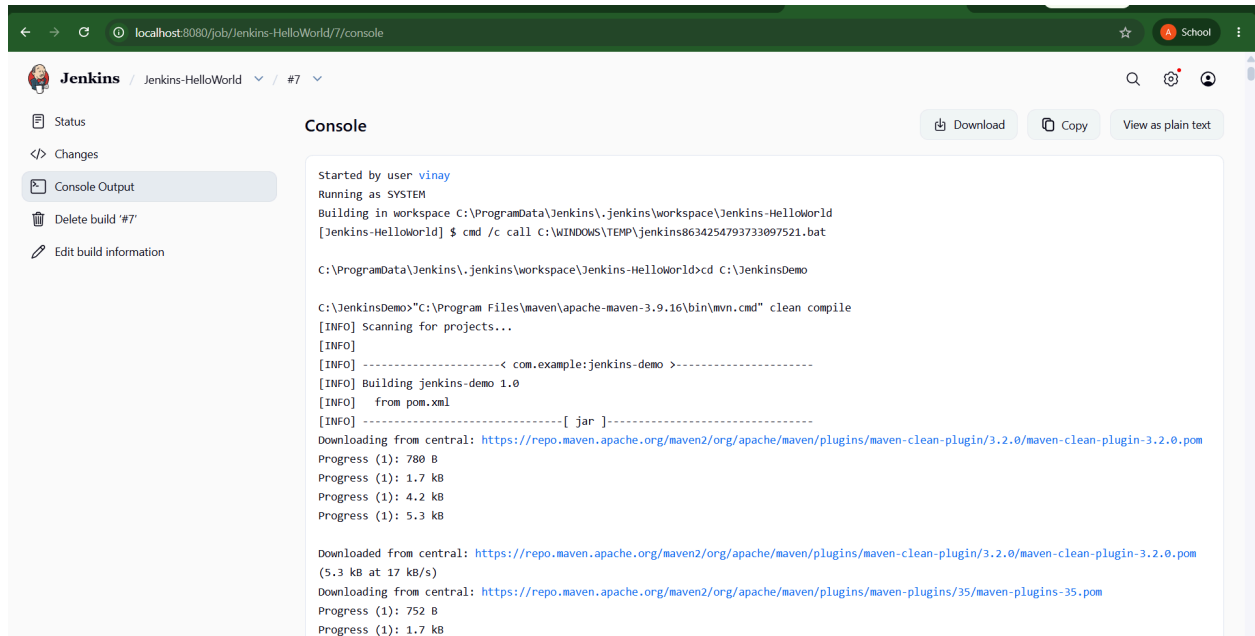


Step 6: See the output

On the left side, you should see:
Build History
#7

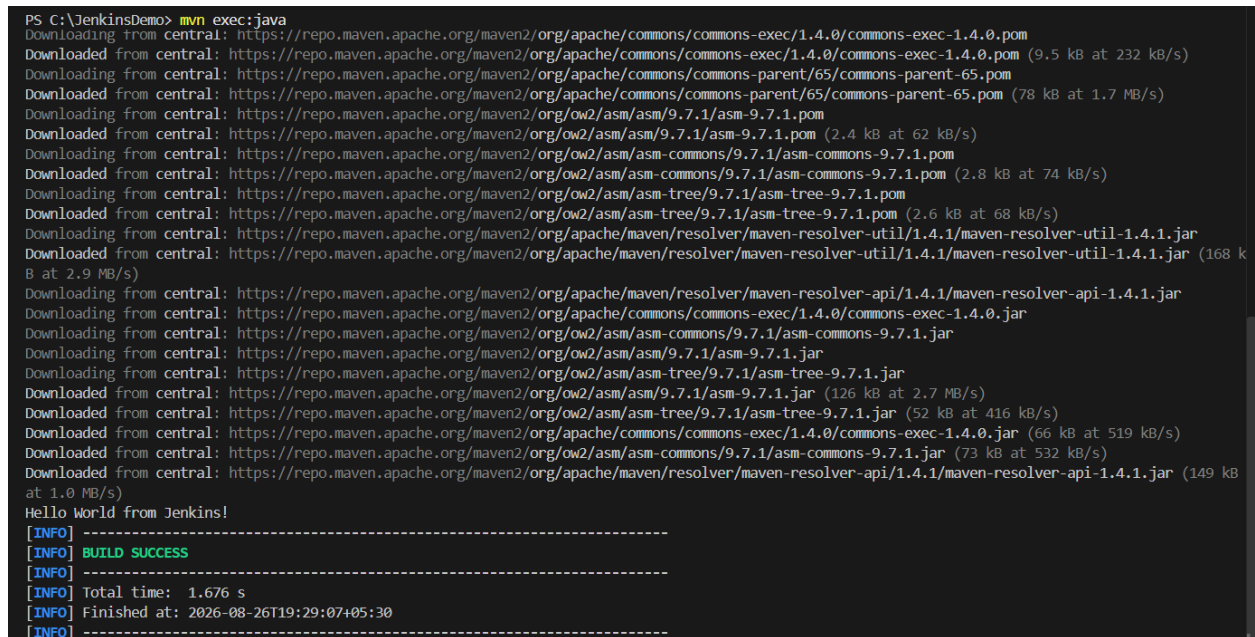


Click:
#1 → Console Output
Look near the bottom.



You should see:

[INFO] BUILD SUCCESS



and:

Hello World from Jenkins!

🎉🎉 Congratulations! You have executed your first Java Maven application using Jenkins.

One thing before you start

Because your pom.xml uses Java 21, make sure Jenkins has access to JDK 21. If you have not yet configured the JDK installation under Manage Jenkins → Tools, tell me what you see

under JDK installations, and I'll give you the exact settings before you create the job.
how to create a folder structure for jenkins demo
Absolutely. Since you are using Windows 10/11, the easiest way is to create the folder structure using File Explorer.